

From Real-World Data to Research Project

A guide to nine datasets for COSMOS projects

Background → Source → Structure → Key Variables → Project Ideas

Goal for Today

Explore each dataset, and consider one that interests you.

Nine Datasets

Data shape	Datasets	Core structure
Multichannel signals	Siena EEG; Dreyer2023 BCI	Recording files, sampling rate, event annotations
Sparse biological matrix	PBMC3K	Genes $\times$ cells, UMI counts, QC metrics
Spatiotemporal tables	Bay Wheels; California SWE	Time + location + measurement
Temporal networks	Copenhagen; Email-Eu; SocioPatterns	Nodes, edges, direction, time, layers
Sports event stream	StatsBomb World Cup	Match $\rightarrow$ event stream $\rightarrow$ players and 360 context

Key idea

Different datasets are not just different sizes; they have different shapes and require different instructions.



Siena Scalp EEG

Clinical scalp EEG recordings with seizure interval annotations

1/9 — Multichannel clinical signal data

Siena Scalp EEG: simulated source observation

Header + waveform samples + annotation

EDF header (one recording)

file: PN00-1.edf

start_time: 19:39:33 duration: 00:43:25

sampling_rate: 512 Hz

channels: Fp1 F3 C3 P3 O1 ... Fz Cz Pz ... EKG1 EKG2

Waveform samples from three channels

sample	time_s	Fp1_uV	F3_uV	C3_uV
0	0.000	-12.4	-8.1	-6.7
1	0.002	-11.8	-7.9	-6.2
2	0.004	-10.9	-7.5	-5.8

Annotation for this recording

seizure_start: 19:58:36 seizure_end: 19:59:46

duration fields locate the interval inside the EDF timeline.

How to read it

- ▶ The header tells us how to interpret the signal array.
- ▶ Each waveform row is one sampled time point for several channels.
- ▶ The seizure annotation gives a time interval, not one label per row.

EEG experiment

[Short video](#)

Simulated source-style observation; numbers are illustrative, but the structure matches the EEG concept.

Siena Scalp EEG: dataset overview

Item	What it means
Source	PhysioNet Siena Scalp EEG Database
Time span	calendar dates are de-identified; each subject has 1–5 EDF files from same or different de-identified days
Participants	14 human subjects undergoing epilepsy monitoring
Raw data object	41 EDF recording files
Main signal type	scalp EEG, with some accompanying physiologic channels
Labels	47 clinically annotated seizure intervals

What to Know

- ▶ This is a **clinical time-series dataset**, NOT a table where one row is one patient.
- ▶ The core object is a **recording file** with many synchronized channels over time.
- ▶ The most important labels are seizure **start** and **end** times within each recording.

Siena Scalp EEG: from data collection to data files

How the data start

Electrodes placed on the scalp measure voltage changes over time. Each electrode contributes one channel signal.

What the file stores

An EDF file stores the continuous recording: header information plus synchronized EEG/EKG signals.

Clinical setting

The dataset comes from epilepsy monitoring: long recordings are reviewed, and seizure intervals are annotated.

Siena Scalp EEG: EDF file and recording variables

What is inside one EDF file?

- ▶ **Header**: start time, duration, channel names, sampling rate, and units.
- ▶ **Signals**: synchronized sample arrays for each channel.
- ▶ **Annotations**: seizure timing information indexed on the recording timeline.

Describe it precisely

- ▶ One **EDF file** = one continuous multichannel recording.
- ▶ One **channel** = one electrode/sensor trace over time.
- ▶ One **sample** = one measurement at one time point in one channel.

Why this matters

Do not describe EDF as “one row = one patient.” For EEG, the main structure is file → channel → time samples, with seizure labels placed on the time axis.

Siena Scalp EEG: 31-channel scalp map vocabulary

How to read channel names

Letters show scalp region: Fp frontopolar, F frontal, C central, P parietal, O occipital, T temporal.

Numbers show side: odd = left, even = right, z = midline.

Example channel groups

Fp: Fp1, Fp2

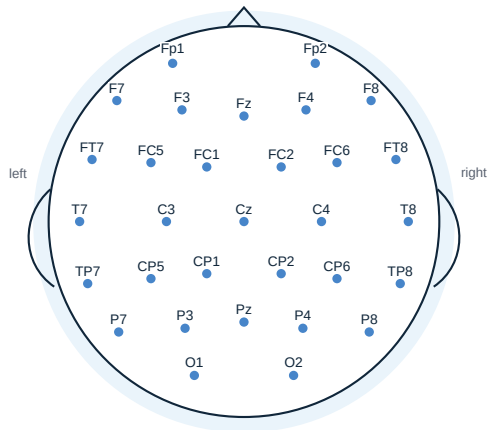
Frontal: F7, F3, Fz, F4, F8

Central: T7, C3, Cz, C4, T8

Posterior: P7, P3, Pz, P4, P8, O1, O2

Why it matters

A channel name tells us **where** on the scalp the signal was recorded.

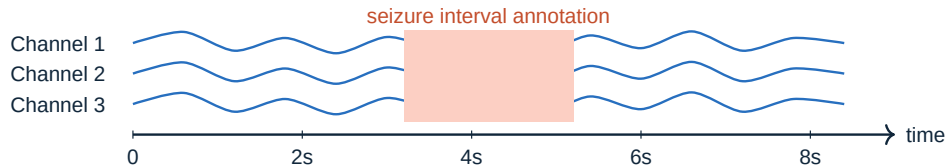


Conceptual 31-channel scalp map; exact channel list should be verified from metadata.

Siena Scalp EEG: essential terminology

Term	Precise meaning
EEG	Electrical activity measured from scalp electrodes over time.
EDF	Multichannel biomedical time-series file: header + synchronized signals.
channel	One electrode or sensor trace, such as EEG or EKG.
sampling rate	512 Hz means 512 samples per second per channel.
seizure interval	Expert-marked start and end times inside a long recording.
interictal / ictal	Outside a seizure / during a seizure.

Siena Scalp EEG: signal and annotation structure



Data-structure point

The label is an interval on the time axis.

Siena Scalp EEG: important variables

Variable	Table / place	Meaning
patient_id	subject metadata	anonymized patient identifier
edf_file	recordings index	one continuous EDF recording
sampling_rate	recordings index	samples per second per channel
duration_seconds	recordings index	recording length in seconds
channel_name	channels index	electrode/sensor label
channel_type	channels index	EEG, EKG, or other channel category
seizure_start_sec	seizures index	start time from recording start
seizure_end_sec	seizures index	end time from recording start

Siena Scalp EEG: scale and limits

Scale item	Value
Subjects	14 human epilepsy patients
Recordings	41 EDF files
Duration	141.02 EDF-header hours
Seizure annotations	47 intervals

Do not assume

- ▶ No pre-filtered waveform windows are exported.
- ▶ A seizure interval is clinical annotation, not a perfect universal truth.
- ▶ Patient metadata should be handled carefully and respectfully.

Siena Scalp EEG: project directions

Good project questions

- ▶ How do channel amplitudes or simple features change around annotated seizures?
- ▶ Can we compare seizure and non-seizure windows after careful window extraction?

First coding steps

- ▶ Read recording and channel indexes.
- ▶ Select one EDF and a small set of EEG channels.
- ▶ Align samples to seizure start/end seconds.
- ▶ Document windowing and filtering choices.

Dreyer2023 BCI

Prompt-based motor-imagery EEG brain-computer-interface
recordings

5/9 — Multichannel experimental signal data

GDF header + event marker + EEG window

GDF header (one participant run)

file: S01_run03.gdf
sampling_rate: 512 Hz
channels: Fz C3 Cz C4 Pz ...

Event marker table

onset_s	event_code	task_label
12.000	769	left hand
20.000	770	right hand

Prompt-aligned EEG window for event at 12.000 s

time_from_prompt_s	C3_uV	Cz_uV	C4_uV
0.000	4.2	1.5	-2.0
0.002	4.6	1.2	-2.4
0.004	4.9	0.9	-2.9

This window becomes one trial: label = left hand.

How to read it

- ▶ The GDF stores continuous EEG plus event markers.
- ▶ A task label comes from the marker code, not from looking at the waveform later.
- ▶ Each analysis example is created by cutting a window around a task prompt.

Simulated source-style observation; C3, Cz, and C4 are shown because motor-imagery tasks often focus on central EEG channels.

Dreyer2023 BCI: dataset overview

Item	What it means
Source	Zenodo Dreyer2023 BCI dataset
Time span	calendar dates not stated; each participant completed one single-day recording session
Participants	human subjects performing prompt-based motor-imagery tasks
Raw data object	multichannel EEG recordings with event markers, often stored in GDF files
Main signal type	scalp EEG measured during imagined movement tasks
Labels	trial class labels such as imagined left-hand or right-hand movement

What to Know

- ▶ This is a **trial-based EEG dataset**: signals become examples after you cut them around task prompts.
- ▶ The raw file is continuous, but many analyses focus on short **time windows**.
- ▶ Class labels come from the experiment design, not from visual inspection afterward.

Experimental task

Participants saw task prompts and imagined left-hand or right-hand movement without physically moving.

Data object

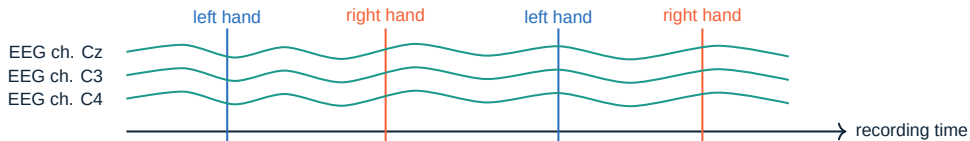
The raw scientific unit is a continuous GDF physiological recording with event annotations.

BCI setting

The goal is to connect brain signals to computer-controlled tasks through structured experimental events.

Term	Precise meaning
BCI	brain-computer interface: using brain signals as computer input
EEG	scalp-recorded electrical activity over time
motor imagery	imagining movement without moving
GDF	file format for physiological recordings and annotations
event	timestamped instruction such as left hand or right hand
OpenViBE	BCI software system used for experimental event annotation
trial window	EEG segment extracted around a task prompt; not automatically exported here

Dreyer2023 BCI: recording structure



Data-structure point

The label is an event code on a continuous signal timeline. Classification requires defining trial windows around events.

Dreyer2023 BCI: GDF files, events, and trial windows

GDF file

A GDF file is a biomedical signal recording format used for EEG/BCI data. It stores channels plus event annotations, similar in spirit to EDF but common in BCI experiments.

Event

A event is a timestamped instruction, such as left-hand or right-hand motor imagery. The label is on the time axis.

Trial window

A trial is not automatically one row in the prepared data. Downstream analysis should cut a time window around each event.

Important

Event code 769 means left-hand event; event code 770 means right-hand event. These are event annotations inside continuous recordings.

Dreyer2023 BCI: important variables

Variable	Meaning
participant_id	human participant identifier
gdf_file	one continuous physiological recording
sampling_rate	512 Hz in the prepared summary
channel_name	signal channel label
channel_type	EEG, EOG, EMG, STIM, or UNKNOWN
event_code	annotation code, e.g., 769 or 770
event_label	mapped meaning, e.g., left_hand or right_hand
onset / time	when the event occurs inside the recording
quality flag	known missing or problematic record indicator

Scale item	Value
Participants	87
GDF recordings	694
Sampling rate	512 Hz
Left-hand events	10,394
Right-hand events	10,398
Quality-flag rows	11

Not included

- ▶ No pre-cut trial tensors.
- ▶ No filtering, artifact rejection, CSP, LDA, or model evaluation.
- ▶ Quality flags are marked, not automatically removed.

Possible questions

Can EEG signals around task prompts distinguish left-hand from right-hand motor imagery?

Project caution

A BCI classification project is not just “train a model.” It requires choices about window length, channels, filtering, artifacts, and evaluation split.

PBMC3K

Single-cell RNA-seq counts from peripheral blood immune cells

2/9 — Sparse biological matrix data

PBMC3K: simulated source observation

Sparse count matrix + row/column metadata

```
matrix.mtx
%%MatrixMarket matrix coordinate integer general
32738 2700 2286884
```

row_index	col_index	UMI_count
1	1	2
1	7	1
14	3	5

```
features.tsv / genes.tsv
1 ENSG00000243485 MIR1302-10
14 ENSG00000198888 MT-ND1
```

```
barcodes.tsv
1 AAACATACAACCAC-1
3 AAACATTGAGCTAC-1
7 AAACATTGATCAGC-1
```

How to read it

- ▶ The matrix only stores nonzero gene-cell counts.
- ▶ Row metadata tells us which gene each row index means.
- ▶ Column metadata tells us which cell barcode each column index means.

Simulated source-style observation; QC metrics are computed later from the count matrix.

PBMC3K: dataset overview

Item	What it means
Source	10x Genomics PBMC 3K
Time span	sample collection date not stated; single time-point healthy-donor sample
Biological material	peripheral blood mononuclear cells (human immune cells)
Raw data object	sparse gene-by-cell count matrix with gene and barcode files
Main measurement	UMI counts for RNA molecules in individual cells
Labels	no official cell-type labels; interpre- tation is mostly downstream

What to Know

- ▶ This is a **matrix dataset**: genes are rows, cells are columns.
- ▶ The key observational unit is often one **cell barcode**, not one patient.
- ▶ Zero-heavy counts are expected, so **sparsity** is part of the data design.

PBMC3K: from blood sample to count matrix

Biological origin

PBMCs are immune cells isolated from blood, including T cells, B cells, NK cells, monocytes, and dendritic cells.

Sequencing idea

Single-cell RNA-seq measures RNA molecules separately for many captured cells.



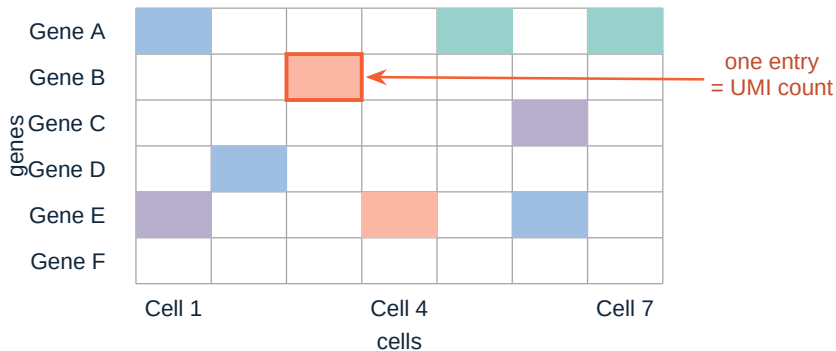
Main object

A sparse gene-by-cell count matrix: rows are genes, columns are cells, entries are UMI counts.

PBMC3K: essential terminology

Term	Precise meaning
PBMC	Peripheral blood mononuclear cell: immune cell type from blood.
single-cell RNA-seq	sequencing method that measures gene expression cell by cell.
gene expression	amount of RNA detected from a gene in a cell.
UMI	unique molecular identifier; helps count original RNA molecules.
barcode	cell identifier assigned during sequencing.
sparse matrix	matrix with many zeros; only nonzero entries are stored efficiently.
QC metric	quality-control variable used to evaluate cell or gene reliability.

PBMC3K: matrix structure



A gene-by-cell matrix is read by rows, columns, and individual count entries.

PBMC3K: important variables and QC metrics

Variable / metric	Level	Meaning
gene id / gene name	gene	identifies a measured gene
cell barcode	cell	identifies one captured cell
UMI count	gene-cell entry	number of molecules counted for that gene-cell pair
total counts	cell QC	total UMIs measured in one cell
detected genes	cell QC	number of genes with nonzero counts in one cell
mitochondrial percent	cell QC	fraction of counts from mitochondrial genes

Why QC matters

A cell with extremely low counts may be failed capture; very high counts may reflect doublets or technical artifacts.

PBMC3K: what is a QC table or QC matrix?

Quality-control table

A QC table is a data-quality summary table. In PBMC3K, a **cell QC table** has one row per cell barcode; a **gene QC table** has one row per gene.

QC matrix

A QC-filtered matrix is the count matrix after removing cells/genes that fail chosen quality rules. It is still genes $\times$ cells, but with fewer rows or columns.

Research habit: QC thresholds are analysis choices; report them before interpreting clusters or marker genes.

QC variable	What it checks
library size / total counts	how much RNA was captured for one cell
detected genes	whether a cell has enough expressed genes
mitochondrial percent	whether a cell may be stressed or low quality
nonzero cells per gene	whether a gene is observed in enough cells

PBMC3K: main files

File type / file	Role
matrix.mtx	raw sparse count matrix
genes.tsv / features.tsv	gene metadata
barcodes.tsv	cell barcode list
processed sparse matrices (.npz)	efficient matrix storage for analysis
cell metadata / QC tables	cell-level summaries and QC metrics
summary / manifest files	dataset counts, provenance, and checks

File-reading habit

For matrix datasets, read the matrix and the row/column metadata together; the matrix alone is not self-explanatory.

PBMC3K: labels, limits, and project directions

Labels and limits

- ▶ No official cell-type labels are included.
- ▶ Clusters or cell types must be inferred downstream.
- ▶ Counts are RNA measurements, not living-cell images.

Good project questions

- ▶ How do QC metrics identify suspicious cells?
- ▶ Can cells be grouped by gene-expression patterns?
- ▶ Which marker genes help interpret clusters?

Bay Wheels San Francisco

Trip-level bike-share records and station-level mobility summaries

3/9 — Urban mobility table data

Bay Wheels: simulated source observation

One trip row from a monthly CSV

```
ride_id,rideable_type,started_at,ended_at,  
start_station_name,end_station_name,start_lat,start_lng,  
end_lat,end_lng,member_casual
```

```
B123,electric_bike,2023-07-14 08:13:20,2023-07-14 08:27:55,  
Market St at 10th St,San Francisco Caltrain Station,  
37.7766,-122.4174,37.7766,-122.3955,member
```

Derived field:
duration_min = 14.58

How to read it

- ▶ One row is one completed trip, not one rider.
- ▶ Start/end timestamps define trip duration.
- ▶ Coordinates describe trip endpoints, not the full route.

Simulated source-style observation; values are illustrative and show the field structure students should inspect.

Bay Wheels: dataset overview

Item	What it means
Source	Bay Wheels System Data
Time span	2024-05-31 to 2024-08-31
Observational unit	one row usually represents one completed bike trip
Raw data object	trip table plus station metadata table
Main measurement	times, durations, origin station, destination station, and rider context
Labels	no formal prediction label; many questions use destination, duration, or traffic patterns as outcomes

What to Know

- ▶ This is a **row-based table dataset**, where each row is a trip event.
- ▶ The most important structure is the link between **time**, **location**, and **trip flow**.
- ▶ Stations act like named places, so station metadata are as important as the trip file.

Bay Wheels: from bike system to trip table

How the data start

A user unlocks a bike, rides, and returns or locks it. The system records start time, end time, endpoint locations, bike type, and rider account type.

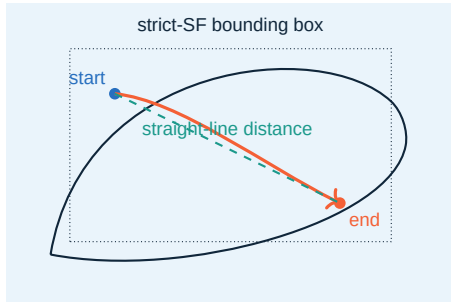
Main data object

One row is one completed trip. Aggregated tables summarize the same trips by station, hour, day, or origin-destination pair.

Bay Wheels: essential terminology

Term	Precise meaning
station	named bike-share location where a trip starts or ends
rideable_type	bike category, such as classic or electric
member_casual	rider account type, not personal identity or demographics
endpoint coordinates	latitude/longitude for start or end, not the route ridden
haversine distance	straight-line distance between endpoints
OD pair	origin-destination station pair, e.g., station A → station B
strict-SF filter	rule keeping trips whose endpoints are inside a San Francisco bounding box

Bay Wheels: OD pair and endpoint-coordinate structure



What the row means

A trip row records the endpoints and timing of a completed ride, not the full GPS path.

Data caution

Do not interpret endpoint coordinates as the exact route. Haversine distance is a straight-line approximation.

Bay Wheels: important variables

Variable	Meaning
ride_id	unique trip identifier
started_at, ended_at	start and end timestamps used to compute duration
start_station_id / name	origin station information
end_station_id / name	destination station information
start_lat, start_lng, end_lat, end_lng	endpoint coordinates
rideable_type	bike category
member_casual	account category of the rider
trip_duration	derived time length of the trip

Bay Wheels: files, scale, and limits

Item	Value / role
Raw files	202406-202408 monthly trip ZIP files
Raw rows	943,556
Clean rows	849,870
Date range	2024-05-31 to 2024-08-31
Processed tables	clean trips, station, day, hour, OD summaries

Not included

- ▶ exact GPS route
- ▶ rider identity
- ▶ weather
- ▶ trip purpose
- ▶ bike inventory history

Bay Wheels: file types, tables, and QC variables

CSV Table meaning

- ▶ Raw monthly trip files are flat tables: one row is one ride record.
- ▶ Aggregate tables will change row meaning: one hour, station, day, or OD pair.

Typical QC checks

- ▶ missing station or coordinate values
- ▶ negative or impossible trip duration
- ▶ endpoints outside the strict San Francisco filter
- ▶ duplicate ride IDs or repeated records

Important distinction

Endpoint coordinates are not GPS traces. They tell where a trip starts and ends, not the route actually ridden.

Bay Wheels: project directions

Possible questions

- ▶ Which stations are most important during commute hours?
- ▶ How do member and casual trips differ in timing or endpoints?
- ▶ Which OD pairs suggest commuting versus tourism patterns?

Project caution

Demand patterns are not automatically caused by weather, events, or policy unless those external data are added and documented.

California SWE

Daily snow water equivalent at California snow stations

4/9 — Environmental station-day data

California SWE: simulated source observation

One station-day row

```
station_id,date,swe_in,snow_depth_in,water_year,  
latitude,longitude,elevation_ft,basin_name
```

```
SIL,2022-03-15,18.4,55.0,2022,  
38.41,-120.22,7100,San Joaquin
```

Station metadata:

```
station_id = SIL  
station_name = Silver Lake  
sensor_type = snow pillow / station measurement
```

How to read it

- ▶ One row is one station observed on one date.
- ▶ SWE is water stored in the snowpack, not snow depth.
- ▶ Basin and elevation help compare stations regionally.

Simulated source-style observation; the key structure is station + date + snow measurement + location metadata.

California SWE: dataset overview

Item	What it means
Source	California Cooperative Snow Surveys
Time span	2011-01-01 to 2020-12-31
Observational unit	one row usually represents one station on one date
Raw data object	station-day table plus station meta-data
Main measurement	snow water equivalent (SWE), date, location, and related snow variables
Labels	no class labels; the target is usually a physical quantity such as SWE or seasonal change

What to Know

- ▶ This is an **environmental time-series table**, organized by station and date.
- ▶ The key concept is that SWE is not snow depth; it is the water stored in the snowpack.
- ▶ The same station is observed repeatedly, so trends and seasonality matter.

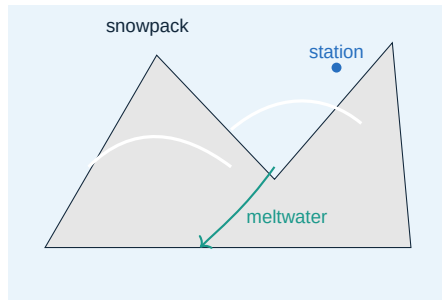
California SWE: from snowpack to station-day table

Real-world system

Mountain snowpack stores water. Snow water equivalent (SWE) estimates how much water would be produced if the snow melted.

Main data object

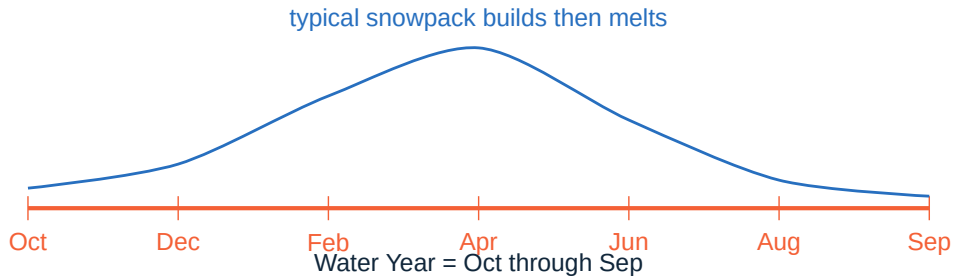
One row is one snow station on one date.



California SWE: essential terminology

Term	Precise meaning
SWE	snow water equivalent: water depth if snowpack melted completely
snow pillow	instrument estimating snow water by measuring snow weight
station-day	one station measured on one date
water year	hydrologic year from Oct 1 to Sep 30
basin	watershed / drainage region used for regional summaries
elevation	station height; important because snow patterns depend strongly on altitude
missing-value code	source values such as -9999 that mean invalid or missing measurement

California SWE: water year and seasonal structure



Why not calendar year?

Snow season starts in one calendar year and melts in the next, so hydrology uses water years.

California SWE: important variables

Variable	Meaning
station_id, station_name	identify the measurement site
date	calendar date of measurement
water_year, month	hydrologic and seasonal time fields
basin_name	watershed or regional grouping
elevation	altitude of station; important for snow accumulation
latitude, longitude	station location
swe_inches	water depth stored in snowpack, in inches
mean_swe_inches	average SWE in an aggregate table

California SWE: files, scale, and limits

Scale item	Value
Clean station-days	412,976
Clean station IDs	131
Basins	26
Date range	2011-01-01 to 2020-12-31
Main table	swe_station_day_clean

Limits

- ▶ It is station data, not a continuous snow map.
- ▶ There is no class label.
- ▶ Missing-value codes must not be treated as real snow measurements.

California SWE: missing values, QC, and station metadata

Station metadata

A station has latitude, longitude, elevation, basin, county or network identifiers, and sometimes sensor type.

QC and missing codes

Source files may use **-9999**, **-999**, or **-99** to mean invalid or missing measurements. These must be converted before analysis.

Why metadata matter

Elevation and basin explain why two stations on the same date can have very different SWE values.

Good documentation

State whether you use raw daily values, cleaned station-days, basin summaries, or water-year summaries.

California SWE: project directions

Possible questions

How does SWE vary by basin, elevation, month, or water year?

Project caution

SWE patterns can suggest hydrologic differences, but drought or climate attribution requires careful external context and additional methods.

Copenhagen Networks

Multi-layer temporal social interaction data

6/9 — Multi-layer network data

Copenhagen Networks: simulated source observation

Same participants, different interaction layers

```
Bluetooth proximity row
user_a,user_b,timestamp,rssi
102,317,2013-09-04 10:35:00,-71

Call row
caller,receiver,timestamp,duration_sec
102,317,2013-09-04 18:12:44,83

SMS row
sender,receiver,timestamp
317,102,2013-09-05 09:02:11

Facebook row
user_a,user_b,friendship
102,317,1
```

How to read it

- ▶ Bluetooth proximity is not the same as communication.
- ▶ Calls and SMS have direction; Facebook and proximity are often treated as undirected.
- ▶ Edge weights are derived by counting events or time bins.

Simulated source-style observation; shown as a compact multi-layer example for the same pair of participants.

Copenhagen Networks: dataset overview

Item	What it means
Source	Figshare Copenhagen Networks Study
Time span	exact calendar dates not stated; four-week smartphone study period
Observational units	participants (nodes) and time-stamped interactions (edges)
Raw data object	multiple network layers such as proximity and communication logs
Main measurement	who interacted with whom, when, and in which layer

What to Know

- ▶ This is a **multi-layer network dataset**, not just one spreadsheet.
- ▶ A **node** is a participant; an **edge** is an interaction.
- ▶ Different layers capture different meanings: physical proximity is not the same as communication.

Copenhagen Networks: from smartphones to layers

Data collection

Participants carried smartphones that logged Bluetooth proximity, phone calls, SMS metadata, and Facebook friendship.

Data shape

Same participants, different interaction layers. Each layer has its own edge meaning and direction.

Copenhagen Networks: node, edge, layer, and matrix vocabulary

Network object

- ▶ **Node**: one participant.
- ▶ **Edge**: one interaction or relationship between two participants.
- ▶ **Layer**: one interaction type: Bluetooth, calls, SMS, or Facebook.

Matrix object

- ▶ **Adjacency matrix**: row and column are participants.
- ▶ **$A[i,j]$** : weight from participant i to participant j .
- ▶ **Sparse matrix**: compact storage for mostly zero connections.

Research habit

Do not mix layer meanings: Bluetooth proximity, phone calls, SMS, and Facebook friendship measure different social signals.

Copenhagen Networks: layer vocabulary

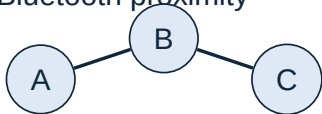
Layer	Direction	Weight meaning
Bluetooth	undirected	count of 5-minute pair bins with proximity contact
Calls	directed	call count; duration stored separately
SMS	directed	text-message count
Facebook	undirected	friendship indicator
Metadata	node-level	gender metadata and node index maps

Important distinction

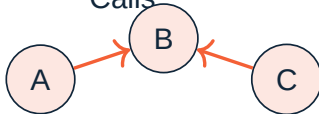
Bluetooth proximity is not friendship. Call/SMS metadata are not message content.

Copenhagen Networks: multi-layer structure

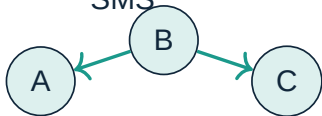
Bluetooth proximity



Calls



SMS



Facebook friendship



Same participants can appear in different layers, but each layer has a different edge meaning.

Copenhagen Networks: important variables and files

Variable / file	Meaning
user_a, user_b	pair of participants in an interaction layer
timestamp	time of Bluetooth, call, or SMS interaction
RSSI	Bluetooth signal strength; stronger often means closer devices
call duration	length of call event, stored separately from count
A_bt, A_calls, A_sms, A_fb	sparse adjacency matrices for layers
node_ids / node_to_idx	mapping from raw IDs to matrix row/column order
genders_clean	node-level metadata, not automatically a target label

Copenhagen Networks: scale, limits, projects

Scale item	Value
Usable participants	787
Raw Bluetooth rows	5,474,289
Call rows	3,600
SMS rows	24,333
Facebook rows	6,429

Project directions

Compare layers: does physical proximity align with calls, SMS, or Facebook friendship?
Which participants are central in each layer?

Limit

Gender is metadata only.

Email-Eu-Core Temporal

Directed email events in a European research institution

7/9 — Temporal directed network data

Email-Eu-Core: simulated source observation

One temporal directed email edge

```
email-Eu-core-temporal.txt.gz
# src  dst  timestamp_seconds
12    45    248
12    90    251
45    12    560
```

Interpret one row:

```
12 45 248
= user 12 sent one email to user 45
  248 seconds after the observation period began.
```

No email text, subject line, or identity is included.

How to read it

- ▶ One row is one directed email event.
- ▶ Direction matters: source → destination.
- ▶ The timestamp is relative time, not a calendar date.

Simulated source-style observation; the dataset is communication metadata, not message content.

Email-Eu-Core: dataset overview

Item	What it means
Source	SNAP Email-Eu-Core Temporal
Time span	timestamps start at 0; time span is 803 days (no calendar dates released)
Observational unit	one directed email event from sender to receiver at one timestamp
Raw data object	temporal edge list plus node/department metadata
Main measurement	sender, receiver, and time of communication
Labels	department labels for some nodes

What to Know

- ▶ This is a **directed temporal network**: direction and time are both essential.
- ▶ Each row is one communication event, not one person.
- ▶ The same node can behave very differently as a sender and as a receiver.

Email-Eu-Core: from email logs to temporal edges

Data collection

The raw events come from institutional email logs: anonymized sender, anonymized recipient, and timestamp.

Main object

One row is one directed email event: src → dst at a timestamp.

Email-Eu-Core: variables that define a temporal email edge

Variable / concept	Meaning
src	anonymized sender node ID
dst	anonymized receiver node ID
timestamp	seconds from the start of observation; not a calendar date
i, j	internal matrix indices corresponding to src and dst
self-loop	event where src equals dst; not automatically removed
node ID space	full network and department files are separate ID spaces

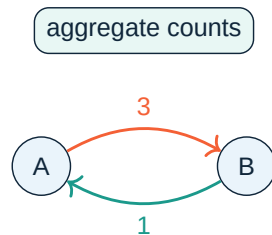
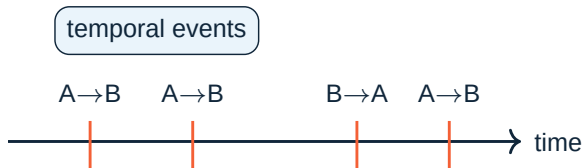
Why this matters

A row is an email event, not an email message body. The dataset contains communication metadata only.

Email-Eu-Core: essential terminology

Term	Precise meaning
directed edge	an email from sender to recipient; direction matters
temporal edge	an event with a timestamp
node ID space	node IDs valid only inside one released file
aggregate edge	repeated emails collapsed into one sender-recipient count
self-loop	edge where sender and recipient are the same node
static graph	graph after collapsing across time

Email-Eu-Core: temporal edge versus aggregate edge



Data-structure point

Temporal edge tables keep each event. Aggregate adjacency matrices count repeated sender-recipient pairs.

Email-Eu-Core: important variables and files

Variable / file	Meaning
src	sender node ID
dst	recipient node ID
timestamp	seconds from start of observation, not calendar date
i, j	internal matrix indexes for src and dst
{key}_edges	clean temporal directed email events
{key}_agg_edges	static sender-recipient counts
{key}_A.npz	sparse directed aggregate adjacency matrix
node_to_idx	raw node ID to matrix index map

Email-Eu-Core: scale, limits, projects

Key	Nodes / temporal edges
full	986 / 327,336
dept1	309 / 60,150
dept2	162 / 46,010
dept3	89 / 12,051
dept4	142 / 47,525

Limits

- ▶ No message text.
- ▶ Department node IDs do not map directly to full-network node IDs.
- ▶ Self-loops are not automatically removed.

Project direction

Study department-level network differences.

SocioPatterns High School

Face-to-face proximity contacts among high-school participants

8/9 — Dynamic contact network data

SocioPatterns: simulated source observation

One 20-second contact row + participant metadata

Contact row

#	t	i	j	class_i	class_j
1385983140	235	451	MP	PC	

Meaning:

participants 235 and 451 were detected in close face-to-face proximity during [t-20s, t].

Metadata rows

participant_id	class	gender
235	MP	F
451	PC	M

How to read it

- ▶ One contact row is a 20-second proximity detection.
- ▶ Class labels describe school-class membership.
- ▶ The row does not prove friendship, conversation, or exact distance.

Simulated source-style observation; the key idea is a temporal contact edge plus node metadata.

SocioPatterns: dataset overview

Item	What it means
Source	SocioPatterns High School Dynamic Contacts
Time span	2011: Dec. 2011 Tue–Fri; 2012: Nov. 2012 Mon–next Tue
Observational unit	one proximity contact recorded during a short time interval
Raw data object	temporal contact edge list plus participant/class metadata
Main measurement	which two people were face-to-face and when
Labels	class membership and participant identifiers

What to Know

- ▶ This is a **dynamic contact network**, recorded in repeated short intervals.
- ▶ A row is an interaction event, not one student profile.
- ▶ Time resolution matters: many small contact rows combine into larger social patterns.

SocioPatterns: from sensors to contact rows

Data collection

Students and teachers wore proximity sensors that detected close face-to-face contacts during school days.

Main object

One row is one pair of participants detected in close proximity during one 20-second interval.

SocioPatterns: contact-row variables and sensor semantics

Variable / term	Meaning
time	endpoint of a 20-second interval; time t represents contact over $[t - 20s, t]$
id_i, id_j	two participants detected in close proximity
class_i, class_j	school class metadata of the two participants
idx_i, idx_j	contiguous internal node indices used for matrices
A_agg[i,j]	total retained contact rows between node i and node j

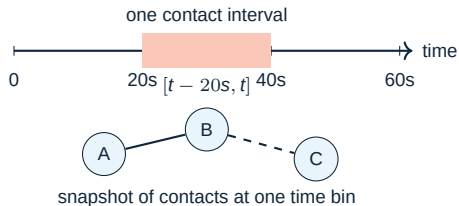
Interpretation warning

A contact row means sensor-detected proximity. It does not prove friendship, conversation, or exact physical distance.

SocioPatterns: essential terminology

Term	Precise meaning
node	a person wearing a proximity sensor
contact edge	two participants detected close together
20-second interval	time window represented by one timestamp endpoint
temporal network	network whose edges occur over time
class label	school class metadata attached to each participant
adjacency matrix	matrix whose entries count contacts between pairs
metadata	participant information such as class and gender

SocioPatterns: 20-second interval structure



Data-structure point

The timestamp is the endpoint. A row at time t represents contact over $[t - 20s, t]$.

Not exact location

The sensors detect close proximity, not GPS position, conversation, or friendship.

SocioPatterns: important variables and files

Variable / file	Meaning
time	interval endpoint timestamp
id_i, id_j	participant pair in a contact row
class_i, class_j	class metadata for participants in a contact row
idx_i, idx_j	internal contiguous node indexes
metadata.csv	one row per participant with class and gender
labels.npy	integer class labels ordered by node IDs
A_agg.npz	aggregate weighted adjacency matrix

SocioPatterns: scale, limits, projects

Year	Nodes	Contacts / snapshots
2011	126	28,560 / 5,609
2012	180	45,047 / 11,273

Class labels: 4 in 2011, 5 in 2012.

Project directions

Study class-to-class mixing.

Limits

Gender is metadata only. Contact means close detection.

StatsBomb 2022 World Cup

Football event streams and 360 freeze-frame context

9/9 — Sports event-stream data

StatsBomb: simulated source observation

One event JSON + one 360 freeze-frame row

```
events/<match_id>.json
{
  "id": "evt_1023",
  "index": 1023,
  "period": 1,
  "timestamp": "00:23:15.420",
  "type": {"name": "Pass"},
  "team": {"name": "Argentina"},
  "player": {"name": "Leo Messi"},
  "location": [45.2, 32.0],
  "pass": {"end_location": [61.0, 28.4], "outcome": null}
}

360 freeze-frame row linked by event_id = evt_1023
player_location=[50.1,34.2], teammate=true, keeper=false
```

How to read it

- ▶ One event is one ordered soccer action inside a match.
- ▶ Nested JSON fields become table columns after flattening.
- ▶ A 360 row describes a visible player around that event, not continuous tracking.

Simulated source-style observation; values are illustrative but match the event-stream and freeze-frame structure.

StatsBomb: dataset overview

Item	What it means
Source	StatsBomb Open Data GitHub
Time span	2022-11-20 to 2022-12-18
Observational units	match-level JSON files containing event rows, lineups, and 360 context frames
Raw data object	nested JSON match files and 360 freeze-frame JSON files
Main measurement	what happened in the match, who did it, where it happened, and nearby player context
Labels	event types, outcomes, teams, players, and tactical context

What to Know

- ▶ This is a **nested event-stream dataset**: one match file contains several related objects.
- ▶ The main unit is often one **event**, but lineups and 360 frames provide extra context.

StatsBomb: from match video to event stream

Data collection

Analysts code a soccer match as an ordered sequence of on-ball events such as passes, carries, shots, pressures, and fouls.

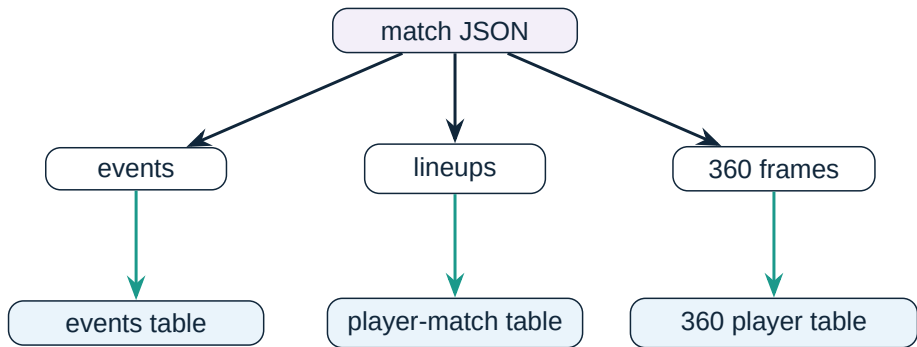
Main object

One row in the event table is one ordered soccer event inside one match.

StatsBomb: essential terminology

Term	Precise meaning
event stream	ordered list of soccer actions inside a match
event_id	unique identifier used to connect event-specific tables
pitch coordinates	StatsBomb field coordinate system on a 120 by 80 pitch
pass / carry / shot	common event types stored as event-specific subsets
xG	expected goals: estimated chance that a shot becomes a goal
possession	sequence of events where one team is considered to control the ball
360 freeze-frame	visible-player context around selected events; not continuous tracking
lineups	match roster: which players are available/participating for each team

StatsBomb: JSON tree to processed tables



Data-structure point

Raw JSON is nested; preprocessing flattens it into tables for analysis.

StatsBomb: JSON, lineups, and table joins

JSON structure

Raw StatsBomb files are nested JSON objects: match metadata, lineups, events, and 360 frames are stored as connected structured objects.

Join keys

Use identifiers such as **match_id**, **event_id**, **team_id**, and **player_id** to connect events, shots, passes, lineups, and 360 tables.

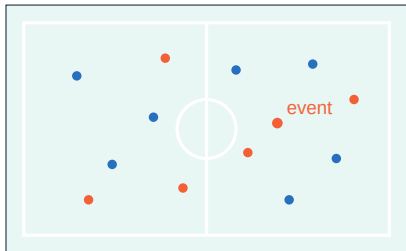
Lineups

Lineups are the cast list of a match: players on each team, IDs, jersey numbers, positions, and lineup metadata.

Important

Nested JSON is often the source format; Parquet tables are the analysis format after flattening.

StatsBomb: event and 360 structure



one event + visible players in a freeze frame

Freeze frame

One event can have many visible-player rows, each with location and context variables.

Not tracking

StatsBomb 360 is not fixed-frequency tracking of every player.

StatsBomb: 360 freeze-frame variables

Variable / term	Meaning
event_id	connects a 360 frame to the event table
visible player row	one player visible in one event-level frame
location	player location on the pitch coordinate system
teammate	whether the visible player is on the acting player's team
keeper	whether the visible player is the goalkeeper
freeze frame	event-level snapshot, not fixed-frequency tracking

Do not overclaim

360 data help describe player context around selected events. They do not record every player's continuous movement for the whole match.

StatsBomb: important variables

Variable	Meaning
match_id	identifies one match
event_id	unique ID for connecting event-related tables
event_index	event order inside a match
minute, second	match time information
team_name, player_name	team and player associated with the event
event_type	pass, carry, shot, pressure, etc.
location	pitch coordinate of the event
shot outcome / xG	result and expected-goal value for shots
freeze-frame variables	visible player location, teammate/opponent, goalkeeper status

StatsBomb: scale and limits

Scale item	Value
Matches	64
Teams	32
Events	234,637
Passes	68,515
Carries	53,764
Shots	1,494
360 frames	203,882
Visible-player rows	3,084,800

Limits

- ▶ 360 is not continuous tracking.
- ▶ Coordinates outside plausibility checks may be flagged, not deleted.

StatsBomb: project directions

Possible questions

What events usually relate to shots? How do teams differ in passing or shot locations?

Project caution

Do not treat event data as video tracking.
Define your target carefully: shot outcome, pass completion, or possession value.

A project starts with the curiosity

Minimum project data card

- ▶ Dataset name and a brief description.
- ▶ Main unit of observation.
- ▶ Key files and how they connect.
- ▶ Important variables and units.
- ▶ Label structure or statement that no label is provided.
- ▶ QC variables, missing values, flags, and exclusions.

Good projects begin with curiosity and a clear understanding of the data.